MCU Lab3 Hard

Testcase 1

Code Template

```
LIST    P=18F4520
#include <p18f4520.inc>
CONFIG  OSC = INTIO67
CONFIG  WDT = OFF

        ORG     0x0000
        GOTO    start

        ORG     0x0020
start:
;-----------------------------------------------------------------------
;       Instruction Start
;-----------------------------------------------------------------------
        MOVLW   0x04
        MOVWF   0x00, ACCESS
        MOVLW   0x05
        MOVWF   0x01, ACCESS
        MOVLW   0x03
        MOVWF   0x02, ACCESS
        MOVLW   0x01
        MOVWF   0x03, ACCESS
        MOVLW   0x02
        MOVWF   0x04, ACCESS
        MOVLW   0x00
        MOVWF   0x05, ACCESS
;-----------------------------------------------------------------------
;       Instruction End
;-----------------------------------------------------------------------

; Enter your code here
```

Explain

- PC = 0x00, Instruction: a=0x04, b=0x05, c=0x03
  Set [0x05] to [0x05] - [0x04] = 0x00 - 0x02 = 0xFE, jump to 0x03
- PC = 0x03, Instruction: a=0x01, b=0x02, c=0xFE
  Set [0x02] to [0x02] - [0x01] = 0x03 - 0x05 = 0xFE, jump to 0xFE

- PC = 0xFE, terminate

Result:

[0x05] = 0xFE, [0x02] = 0xFE

Testcase 2

Code Template

```
LIST    P=18F4520
#include <p18f4520.inc>
CONFIG  OSC = INTIO67
CONFIG  WDT = OFF

        ORG     0x0000
        GOTO    start

        ORG     0x0020
start:
;-------------------------------------------------------------------------
;       Instruction Start
;-------------------------------------------------------------------------
        MOVLW   0x09
        MOVWF   0x00, ACCESS
        MOVLW   0x0C
        MOVWF   0x01, ACCESS
        MOVLW   0x03
        MOVWF   0x02, ACCESS
        MOVLW   0x0A
        MOVWF   0x03, ACCESS
        MOVLW   0x0B
        MOVWF   0x04, ACCESS
        MOVLW   0xFE
        MOVWF   0x05, ACCESS
        MOVLW   0x0D
        MOVWF   0x06, ACCESS
        MOVLW   0x0D
        MOVWF   0x07, ACCESS
        MOVLW   0x00
        MOVWF   0x08, ACCESS
        MOVLW   0xFF
        MOVWF   0x09, ACCESS
        MOVLW   0x01
        MOVWF   0x0A, ACCESS
        MOVLW   0x05
        MOVWF   0x0B, ACCESS
;-------------------------------------------------------------------------
;       Instruction End
;-------------------------------------------------------------------------

; Enter your code here
```

(https://hackmd.io/@rota1001) / 微算機 Lab3 Hard

Explain

- PC = 0x00, Instruction: a=0x09, b=0x0C, c=0x03
  Set [0x0C] to [0x0C] - [0x09] = 0x00 - 0xFF = 0x01
- PC = 0x03, Instruction: a=0x0A, b=0x0B, c=0xFE
  Set [0x0B] to [0x0B] - [0x0A] = 0x05 - 0x01 = 0x04
- PC = 0x06, Instruction: a=0x0D, b=0x0D, c=0x00
  Set [0x0D] to [0x0D] - [0x0D] = 0x00 - 0x00 = 0x00, jump to 0x00
- PC = 0x00, Instruction: a=0x09, b=0x0C, c=0x03
  Set [0x0C] to [0x0C] - [0x09] = 0x01 - 0xFF = 0x02
- PC = 0x03, Instruction: a=0x0A, b=0x0B, c=0xFE
  Set [0x0B] to [0x0B] - [0x0A] = 0x04 - 0x01 = 0x03
- PC=0x06, Instruction: a=0x0D, b=0x0D, c=0x00
  Set [0x0D] to [0x0D] - [0x0D] = 0x00, jump to 0x00
  ...
- [0x0C] = 0x04, [0x0B] = 0x01
- PC=0x00, Instruction: a=0x09, b=0x0C, c=0x03
  Set [0x0C] to [0x0C] - [0x09] = 0x04 - 0xFF = 0x05
- PC=0x03, Instruction: a=0x0A, b=0x0B, c=0xFE
  Set [0x0B] to [0x0B] - [0x0A] = 0x01 - 0x01 = 0x00, jump to 0xFE
- PC=0xFE, terminate

Result:
[0x0C] = 0x05, [0x0B] = 0x00, [0x0D] = 0x00